Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic sequences: finding terms**

Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.

$a_{12} = 5 + 11(3) = 5 + 33 = 38.$

2 **[Foundational]** • **Target: Arithmetic sequences: finding terms**

Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.

$a_{20} = -4 + 19(2) = -4 + 38 = 34.$

3 **[Foundational]** • **Target: Arithmetic sequences: finding terms**

Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.

$a_{15} = 100 + 14(-6) = 100 - 84 = 16.$

4 **[Foundational]** • **Target: Arithmetic sequences: finding terms**

Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.

$7,\ 5,\ 3,\ 1.$

5 **[Foundational]** • **Target: Arithmetic sequences: finding a and d**

The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d.

$a + 3d = 17$ and $a + 8d = 42$. Subtracting: $5d = 25 \Rightarrow d = 5$. Then $a = 17 - 15 = 2$.

6 **[Foundational]** • **Target: Arithmetic sequences: finding a and d**

An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d.

$a_6 = a_1 + 5d = 10 + 5d = 30 \Rightarrow 5d = 20 \Rightarrow d = 4.$

7 **[Foundational]** • **Target: Arithmetic series: the sum formula**

Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using $S_n = \frac{n}{2}(2a + (n-1)d)$.

$S_{20} = \frac{20}{2}(6 + 19(4)) = 10(6 + 76) = 10(82) = 820.$

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.

$S_{15} = \frac{15}{2}(100 + 14(-3)) = 7.5(100 - 42) = 7.5(58) = 435$.

9 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the arithmetic series $2 + 5 + 8 + \cdots + 50$.

Here $a = 2$, $d = 3$; $50 = 2 + (n-1)3 \Rightarrow n = 17$. $S_{17} = \frac{17}{2}(2 + 50) = 17(26) = 442$.

10 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 10 positive multiples of 6.

These form an arithmetic sequence with $a = 6$, $d = 6$, $n = 10$.

$S_{10} = \frac{10}{2}(12 + 9(6)) = 5(12 + 54) = 5(66) = 330$.

11 [Intermediate] • Target: Geometric sequences: finding terms

Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

$a_6 = 3(2)^5 = 3(32) = 96$.

12 [Intermediate] • Target: Geometric sequences: finding terms

Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

$a_4 = 2(-3)^3 = 2(-27) = -54$.

13 [Intermediate] • Target: Geometric sequences: finding terms

Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.

$5,\ 15,\ 45,\ 135$.

14 [Intermediate] • Target: Geometric sequences: finding a and r

The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a.

$ar = 6$ and $ar^4 = 162$. Dividing: $r^3 = 27 \Rightarrow r = 3$. Then $a = 6/3 = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 1.01 Algebra & Equations

A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r.

$a_4 = a_1 r^3 = 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

16 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 2.01 Functions & Transformations

A geometric sequence has 3rd term 12 and 6th term 96. Find a and r.

$ar^2 = 12$ and $ar^5 = 96$. Dividing: $r^3 = 8 \Rightarrow r = 2$. Then $a = 12/4 = 3$.

17 [Challenging] • Target: Geometric series: the finite sum • Cross-Lesson: 4.01 Trigonometry

Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

$S_6 = \frac{3(2^6 - 1)}{2 - 1} = 3(63) = 189$.

18 [Challenging] • Target: Geometric series: the finite sum • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the sum of the first 5 terms of the geometric series with $a = 100$, $r = 0.5$.

$S_5 = \frac{100(1 - 0.5^5)}{1 - 0.5} = \frac{100(0.96875)}{0.5} = 193.75$.

19 [Challenging] • Target: Geometric series: the finite sum • Cross-Lesson: 6.01 Limits & Derivatives

Find the sum of the first 4 terms of the geometric series with $a = 2$, $r = -3$.

$S_4 = \frac{2(1 - (-3)^4)}{1 - (-3)} = \frac{2(1 - 81)}{4} = \frac{-160}{4} = -40$.

20 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 7.01 Integration & Applications

Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.

$S = \frac{a}{1 - r} = \frac{4}{1 - \frac{1}{2}} = 8$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reading a sequence from its terms

The bar chart shows the first five terms of a sequence. Find the common difference.

a. 6

b. 5 ✓

c. 4

d. 9

Each term increases by 5.

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22 [Foundational] • Target: Reading a sequence from its terms

Using the same bar chart, predict the 8th term.

a. 29

b. 34

c. 39 ✓

d. 44

$d = 5$, $a = 4$: $T_8 = 4 + 7(5) = 39$.

23 [Foundational] • Target: Reading a sequence from its terms

Using the same bar chart, find the sum of the first five terms shown.

a. 70 ✓

b. 66

c. 24

d. 75

$4 + 9 + 14 + 19 + 24 = 70$.

24 [Foundational] • Target: Arithmetic sequences: finding a and d

The 3rd term of an arithmetic sequence is 15 and the 8th term is 45. Find the first term a and common difference d.

a. $a = 3, d = 5$

b. $a = 3, d = 6$ ✓

c. $a = 6, d = 3$

d. $a = 9, d = 6$

$d = \frac{45 - 15}{5} = 6$; $a = 15 - 2(6) = 3$.

25 [Foundational] • Target: Arithmetic series: the sum formula

Find the sum of the first 30 terms of the arithmetic sequence with $a = 1$, $d = 2$ (the odd numbers).

a. 960

b. 930

c. 870

d. 900 ✓

$S_{30} = \frac{30}{2}(2 + 29(2)) = 15(60) = 900$.

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26 [Foundational] • Target: Geometric sequences: finding a and r

A geometric sequence has 2nd term 4 and 5th term -108. Find r.

a. $r = 3$

b. $r = -27$

c. $r = -3$ ✓

d. $r = -9$

$r^3 = -108 \div 4 = -27 \Rightarrow r = -3$.

27 [Foundational] • Target: Geometric sequences: finding a and r

A geometric sequence has $a_1 = 6$ and $a_4 = -162$. Find r.

a. $r = 3$

b. $r = -9$

c. $r = -3$ ✓

d. $r = -27$

$r^3 = -162 \div 6 = -27 \Rightarrow r = -3$.

28 [Foundational] • Target: Infinite geometric series

Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \cdots$.

a. 4.5

b. 9

c. 6

d. 6.75 ✓

$a = 9, r = -\frac{1}{3}$: $S = \frac{9}{1 + \frac{1}{3}} = \frac{9}{4/3} = 6.75$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Infinite geometric series

Express the repeating decimal $0.777\ldots$ as a fraction using an infinite geometric series with $a = 0.7$, $r = 0.1$.

a. $\frac{7}{90}$

b. $\frac{77}{100}$

c. $\frac{7}{9}$ ✓

d. $\frac{7}{10}$

$S = \frac{0.7}{1 - 0.1} = \frac{0.7}{0.9} = \frac{7}{9}$.

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30 [Intermediate] · Target: Sigma notation

Evaluate $\sum_{k=1}^{6}(3k-1)$.

a. 51

b. 54

c. 57 ✓

d. 63

$2+5+8+11+14+17=57$.

31 [Intermediate] · Target: Sigma notation

Evaluate $\sum_{k=2}^{5}k^2$.

a. 55

b. 30

c. 50

d. 54 ✓

$4+9+16+25=54$.

32 [Intermediate] • Target: Recursive sequences

A sequence satisfies $a_1=2$ and $a_{n+1}=2a_n+1$. Find a_4.

a. 11

b. 23 ✓

c. 5

d. 47

$2, 5, 11, 23$: $a_4=23$.

33 [Intermediate] • Target: Identifying arithmetic and geometric sequences

State whether $6, 11, 16, 21, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

a. **Arithmetic,** $d=5$ ✓

b. Geometric, $r=\frac{11}{6}$

c. Geometric, $r=5$

d. Arithmetic, $d=6$

Each term adds 5: arithmetic, $d=5$.

34 [Intermediate] • Target: Identifying arithmetic and geometric sequences

State whether $90, 80, 70, 60, \ldots$ is arithmetic or geometric, and find its 12th term.

a. **Arithmetic,** -20 ✓

b. Arithmetic, -10

c. Arithmetic, 30

d. Geometric, -20

$d=-10$: $T_{12}=90+11(-10)=-20$.

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35 [Intermediate] • Target: Word problems: sequences in context

An employee's starting monthly salary is 3,500 dollars, and it increases by 150 dollars each year. Find the salary in year 6.

a. 3,950 dollars

b. 4,400 dollars

c. 4,100 dollars

d. 4,250 **dollars** ✓

$a = 3{,}500$, $d = 150$: $T_6 = 3{,}500 + 5(150) = 4{,}250$.

36 [Intermediate] • Target: Word problems: sequences in context

A bacteria culture starts with 300 cells and doubles every hour. Find the number of cells after 5 hours.

a. 19,200

b. 1,500

c. 9,600 ✓

d. 4,800

$300 \times 2^5 = 300 \times 32 = 9{,}600$.

37 [Intermediate] • Target: Word problems: sequences in context

A theater's ticket sales grow according to a geometric sequence: 200 tickets sold on day 1, with a common ratio of 1.5. Find the number of tickets sold on day 4.

a. 900

b. 675 ✓

c. 300

d. 450

$T_4 = 200(1.5)^3 = 200(3.375) = 675$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Sequences & Series • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the 5th term of the arithmetic sequence with $a = -19$, $d = -5$.

a. -36

b. -39 ✓

c. -44

d. -34

$a_5 = -19 + 4((-5)) = -19 - 20 = -39$.

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39 [Challenging] • Target: Sequences & Series • Cross-Lesson: 6.01 Limits & Derivatives

Find the 7th term of the geometric sequence with $a = -2$, $r = 2$.

a. -131

b. -128 ✓

c. 128

d. -125

$a_7 = -2(2)^6 = -2(64) = -128$.

40 [Challenging] • Target: Sequences & Series • Cross-Lesson: 7.01 Integration & Applications

Find the 3rd term of the geometric sequence with $a = -2$, $r = 2$.

a. -8 ✓

b. 8

c. -11

d. -5

$a_3 = -2(2)^2 = -2(4) = -8$.

41 [Challenging] • Target: Sequences & Series • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the 6th term of the geometric sequence with $a = -9$, $r = -2$.

a. 291

b. -288

c. 285

d. 288 ✓

$a_6 = -9((-2))^5 = -9((-32)) = 288$.

42 [Challenging] • Target: Sequences & Series • Cross-Lesson: 9.01 Statistics & Probability

Find the 3rd term of the geometric sequence with $a = -5$, $r = -3$.

a. -48

b. -42

c. -45 ✓

d. 45

$a_3 = -5((-3))^2 = -5(9) = -45$.

43 [Challenging] • Target: Sequences & Series • Cross-Lesson: 1.01 Algebra & Equations

Evaluate $\sum_{k=1}^{5} (k + 4)$.

a. 31

b. -35

c. 39

d. 35 ✓

Terms: $5, 6, 7, 8, 9$. Sum $= 35$.

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44 [Challenging] • Target: Sequences & Series • Cross-Lesson: 2.01 Functions & Transformations

Evaluate $\sum_{k=1}^{6} (k-2)$.

a. -9

b. 13

c. 9 ✓

d. 5

Terms: $-1, 0, 1, 2, 3, 4$. Sum $= 9$.

45 [Challenging] • Target: Sequences & Series • Cross-Lesson: 4.01 Trigonometry

A sequence satisfies $a_1 = 6$ and $a_{n+1} = a_n + 2$. Find a_6.

a. 14

b. 13

c. 16 ✓

d. 19

$a_1 = 6, a_2 = 8, a_3 = 10, a_4 = 12, a_5 = 14, a_6 = 16$.

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